CODEFORCES
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Codeforces Round 1133 (Div. 1, Div. 2) is coming soon! Register here.

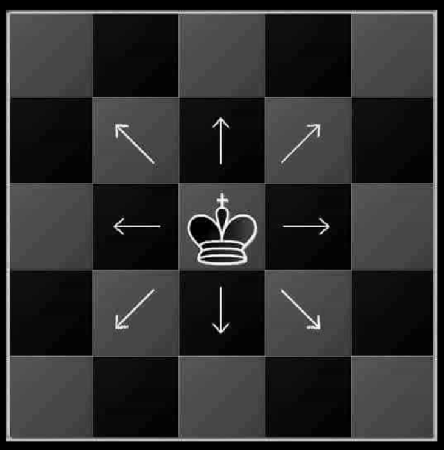
PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

F. The King of Königsberg

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given an $N * N$ chessboard and the initial position (x, y) of a King piece. The King can move one step in any of the 8 possible directions: Up, Down, Left, Right, Top-left diagonal, Top-right diagonal, Bottom-left diagonal, Bottom-right diagonal.



Moves of a King in Chess

Your task is to determine the number of valid moves the King can make in one move. A move is valid if it remains inside the board.

Input

The first line contains an integer $(1 \leq N \leq 2 \times 10^5)$ — the size of the chessboard.

The second line contains two integers $(1 \leq x, y \leq N)$ — the initial position of the King on the chessboard.

Output

First, print an integer K — the number valid moves the King can make in one move.

Next, print K lines, each containing two integers representing a valid move in ascending order. A move (a, b) is smaller than (c, d) if $a < c$ or if $a = c$ and $b < d$.

Examples

input	Copy
8 1 1	

output	Copy
3 1 2	

CSE221 Assignment 04 Fall 2026
Contest is running
8 days
Contestant

→ Submit?

Language: Java 21 64bit

Choose file: Choose File No file chosen

Verify you are human

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Submit

→ Last submissions

Submission	Time	Verdict
397686393	Nov/07/2026 09:10	Accepted

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2 1
2 2
```

input Copy

```
8
1 2
```

output Copy

```
5
1 1
1 3
2 1
2 2
2 3
```

input Copy

```
8
2 2
```

output Copy

```
8
1 1
1 2
1 3
2 1
2 3
3 1
3 2
3 3
```

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The only programming contests Web 2.0 platform
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